

R 's complement

r 's complement

(radix complement)

$(r-1)$'s complement

(Diminished Radix complement)

e) complement of $(7)_{10}$

$$\text{Ans} = 10 - 7 = 3$$

$$(9)_{10} \quad 10 - 9 = 1$$

$$(6)_{10} \quad 10 - 6 = 4$$

$\therefore r^n - N$ Formula for r 's complement

(digit)

$$N = 7, r = 10, n = 1$$

8) $N = 5690$, determine 10 's complement

$$r = 10, N = 5690, n = 4 \text{ digits} = 4$$

$$10^4 - 5690$$

$$= 10000 - 5690$$

$$= 4310$$

9) $(58)_{10}$, find R 's complement

$$N = 58, r = 10, n = 2$$

$$r^n - N$$

$$= 10^2 - 58$$

$$= 42$$

10) $N = 1101$, find 2 's complement

$$r = 2, N = 1101, n = 4$$

$$= 2^4 - 1101$$

$$= 16 - 1101 \quad \therefore 16 \text{ in binary} = 10000$$

$$= 10000 - 1101$$

$$\begin{array}{r} 011010 \\ \hline 00011 \end{array}$$

$$= 11$$

$(r-1)$'s complement

$$(r-1)'s \text{ complement} = \boxed{r^n - 1} - 1$$
$$= r's \text{ complement} - 1$$

6) $(5674)_9$ find 7's complement

$$8^4 = 5674 \xrightarrow{\text{2's octal}}$$

$$= (4076)_{10} \xrightarrow{\text{convert to decimal}} = 5674 - 1$$

$$= (4095)_{10}$$

$$= 7777 - 5674$$

$$= 2103$$

